

Network and discrete optimization

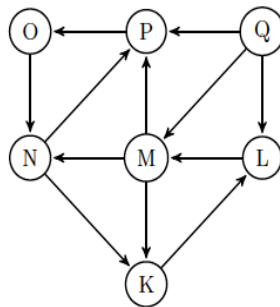
Michel Bierlaire

Graded quiz

1 Networks

Question 1

In the following directed graph, there is **no** forward path from node O to the node ...



Solution 1: L

Solution 2: M

Solution 3: P

Solution 4: Q

Question 2

Let G be a directed graph with m nodes and n arcs. If G is a tree, which one of the following statements is **False**?

Solution 1: G has no cycle and a simple cycle is formed if any arc is added.

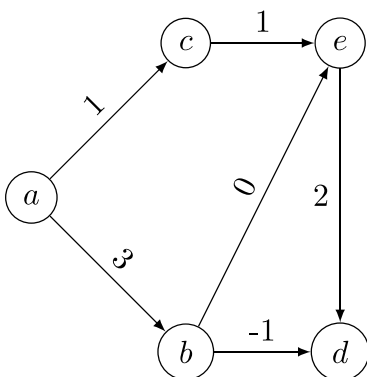
Solution 2: G is connected and $n = m - 1$.

Solution 3: G is connected and the removal of any single arc disconnects the graph.

Solution 4: There is a unique simple forward path connecting any two nodes.

Question 3

Consider the following network where the number associated with each arc represents its costs.



The incidence matrix of this network is:

Solution 1:
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{pmatrix}$$

Solution 2:
$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

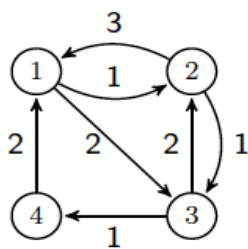
Solution 3:
$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Solution 4:

$$\begin{pmatrix} 0 & 1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Question 4

Which adjacency matrix could represent this graph where the number associated with each arc represents its costs?



Solution 1:
$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Solution 2:
$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 3 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 2 & 0 & 0 & 0 \end{pmatrix}$$

Solution 3:
$$\begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Solution 4:
$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 3 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}$$

2 Transshipment

Question 5

Consider a network $(\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ is the set of nodes and $\mathcal{A}$ is the set of arcs. If the supply or demand of each node is given as vector $s_i, i \in \mathcal{N}$, the cost of transporting a unit of flow on each arc as $c_{ij}, (i, j) \in \mathcal{A}$, while $x_{ij}, (i, j) \in \mathcal{A}$ are the decision variables determining the flow on each arc (i, j) , which of the following is a correct formulation of the transshipment problem?

Solution 1:

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} c_{ij} x_{ij} \\ & \text{s.t.} \\ & \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N} \\ & x_{ij} \geq 0, \forall (i, j) \in \mathcal{A} \end{aligned}$$

Solution 2:

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} \\ & \text{s.t.} \\ & \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N} \\ & x_{ij} \geq 0, \forall (i, j) \in \mathcal{A} \end{aligned}$$

Solution 3:

$$\begin{aligned} & \max_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} \\ & \text{s.t.} \\ & \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N} \\ & x_{ij} \geq 0, \forall (i, j) \in \mathcal{A} \end{aligned}$$

Solution 4:

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = 0, \forall i \in \mathcal{N} \\ & x_{ij} \geq 0, \forall (i,j) \in \mathcal{A} \end{aligned}$$

Question 6

Consider the following primal linear optimization problem:

$$\begin{aligned} & \max 3x_1 + 4x_2 \\ \text{s.t:} \\ & x_1 + 2x_2 \leq 10, \\ & x_1 + x_2 \leq 8, \\ & 3x_1 + 5x_2 \leq 40, \\ & x_1, x_2 \geq 0. \end{aligned}$$

The optimal solution of the primal is $x^* = (6, 2)$. By using the complementary slackness conditions, determine which one of the following solution is the optimal solution of the dual.

Solution 1: $y^* = (1, 2, 0)$.

Solution 2: $y^* = (\frac{1}{5}, 3, 0)$

Solution 3: $y^* = (\frac{3}{5}, 0, \frac{1}{2})$

Solution 4: $y^* = (0, 2, \frac{1}{4})$

Question 7

The total unimodularity of the incidence matrix of a transshipment problem allows us to

Solution 1: guarantee that the solution of the transshipment problem are integer, if the data is integer.

Solution 2: reduce the number of variables of the problem.

Solution 3: express the transshipment problem as a shortest path problem.

Solution 4: guarantee that the problem is feasible.

Question 8

Which of the following statements is true ?

Solution 1: For a longest path problem in an acyclic network, the cost of the longest path is integer if all costs are integer.

Solution 2: If the incidence matrix is totally unimodular, the optimal solution of a transshipment problem is integer regardless of the values of the supply vector s .

Solution 3: An optimal solution to an integer linear optimization problem exists only if the matrix A is totally unimodular.

Solution 4: If the optimal value of the objective function of a linear program is integer, then the matrix A is totally unimodular.

3 Shortest path

Question 9

Consider the connected graph $G(m, n)$ with m the number of nodes and n the number of arcs. What statement is **FALSE**?

Solution 1: The number of arcs in Bellman's subnetwork is always $m - 1$.

Solution 2: Bellman's subnetwork is always a tree.

Solution 3: If the graph $G(m, n)$ contains a negative cost cycle then there is no Bellman's subnetwork.

Solution 4: The shortest path algorithm terminates after a finite number of iterations.

Question 10

Consider Dijkstra's algorithm to calculate the shortest paths from a node to all the other nodes. Which one from the following conditions is sufficient for the labels to be permanent?

Solution 1: The cost of all arcs must be non-negative.

Solution 2: The network must not contain a negative cost cycle.

Solution 3: The rank of the origin node must be greater than 1.

Solution 4: The arc costs are integer.

Question 11

Consider Dijkstra's algorithm for finding the shortest path between two nodes in a graph with negative costs. Which of the following statements regarding the convergence of the algorithm is true?

Solution 1: The algorithm will perform at most as many iterations as there are nodes in the graph.

Solution 2: If no cycle with negative costs exists, the algorithm will terminate in a finite number of steps.

Solution 3: The algorithm will always terminate in a finite number of steps.

Solution 4: The algorithm can stop once the destination node has been visited.

Question 12

Consider a network $(\mathcal{N}, \mathcal{A})$ of m nodes and n arcs, cost vector $c \in \mathbb{R}^n$, and the origin node o . Dijkstra's algorithm has the property that each node is processed at most once by the algorithm. Under which condition is this true?

Solution 1: The network must be connected.

Solution 2: The number of nodes must be strictly less than the number of arcs.

Solution 3: The network must not contain cycles.

Solution 4: The cost vector c must be non-negative.

4 Discrete optimization

Question 13

Suppose that we have three functions $f(i)$, $g(i)$ and $h(i)$, $i \in \mathcal{A} \subset \mathbb{N}$ ($\mathcal{A}$ is an arbitrary subset of $\mathbb{N}$), such that for all i we have $0 \leq f(i) \leq \mathcal{M}_f$, $0 \leq g(i) \leq \mathcal{M}_g$ and $0 \leq h(i) \leq \mathcal{M}_h$. We would like to model that for each i , at least one of the three functions must be equal to 0. This can be modeled as:

Solution 1:

$$\begin{aligned} a_{fi}f(i) &\leq \mathcal{M}_f, \forall i \in \mathcal{A} \\ a_{gi}g(i) &\leq \mathcal{M}_g, \forall i \in \mathcal{A} \\ a_{hi}h(i) &\leq \mathcal{M}_h, \forall i \in \mathcal{A} \\ a_{fi} + a_{gi} + a_{hi} &= 1, \forall i \in \mathcal{A} \\ a_{fi}, a_{gi}, a_{hi} &\in \{0, 1\}, \forall i \in \mathcal{A} \end{aligned}$$

Solution 2:

$$\begin{aligned} f(i) &\leq a_{fi}\mathcal{M}_f, \forall i \in \mathcal{A} \\ g(i) &\leq a_{gi}\mathcal{M}_g, \forall i \in \mathcal{A} \\ h(i) &\leq a_{hi}\mathcal{M}_h, \forall i \in \mathcal{A} \\ a_{fi} + a_{gi} + a_{hi} &= 1, \forall i \in \mathcal{A} \\ a_{fi}, a_{gi}, a_{hi} &\in \{0, 1\}, \forall i \in \mathcal{A} \end{aligned}$$

Solution 3:

$$f(i) + g(i) + h(i) = 1, \forall i \in \mathcal{A}$$

Solution 4:

$$f(i) \leq a_{fi} \mathcal{M}_f, \forall i \in \mathcal{A}$$

$$g(i) \leq a_{gi} \mathcal{M}_g, \forall i \in \mathcal{A}$$

$$h(i) \leq a_{hi} \mathcal{M}_h, \forall i \in \mathcal{A}$$

$$a_{fi} + a_{gi} + a_{hi} = 1, \forall i \in \mathcal{A}$$

$$a_{fi}, a_{gi}, a_{hi} \in \mathbb{R}, \forall i \in \mathcal{A}$$

Question 14

What linear constraints are equivalent to the following constraints?

$$\begin{aligned}z &= xy, \\ 0 &\leq y \leq M,\end{aligned}$$

where x is a boolean variable, y and z are real variables and M is a constant.

Solution 1:

$$\begin{aligned}x + y &\leq M + z, \\ z &\leq y, \\ z &\geq 0.\end{aligned}$$

Solution 2:

$$\begin{aligned}z &\leq Mx, \\ z &\leq y, \\ y - z &\leq M(1 - x), \\ z &\geq 0.\end{aligned}$$

Solution 3:

$$\begin{aligned}z &\leq (1 - x)M + y, \\ (1 - x)M &\leq z, \\ z &\geq 0.\end{aligned}$$

Solution 4:

$$\begin{aligned}z - y &\leq (1 - x)M, \\ z &\leq Mx, \\ z &\geq 0.\end{aligned}$$

Question 15

We are trying to define a binary linear optimization problem whose solution will determine the optimal configuration of a car. The seller imposes that AC and fog lights can be installed together only if at least one of the following items is installed too: side airbags, multimedia or speed limiter. If we consider the following binary variables, that are 1 if the corresponding item is installed, and 0 otherwise: x_{AC} for AC , x_{fl} for fog lights, x_{sa} for side airbags, x_{mm} for multimedia, and x_{sl} for speed limiter, which of the following (sets of) inequalities models the above mentioned constraint?

Solution 1:

$$x_{AC} + x_{fl} - x_{sa} - x_{mm} - x_{sl} \leq 1.$$

Solution 2:

$$\begin{aligned} x_{AC} - x_{sa} - x_{mm} - x_{sl} &\leq 0, \\ x_{fl} - x_{sa} - x_{mm} - x_{sl} &\leq 0. \end{aligned}$$

Solution 3:

$$x_{AC}x_{fl} - x_{sa} - x_{mm} - x_{sl} \leq 0.$$

Solution 4:

$$x_{AC} + x_{fl} - x_{sa} - x_{mm} - x_{sl} \leq 2.$$

Question 16

Consider a linear integer minimization problem P and its relaxation $R(P)$. Which one of the following statements is **wrong** ?

Solution 1: An upper bound on the optimal solution of P can be obtained by rounding up the optimal solution of $R(P)$.

Solution 2: The problem $R(P)$ is the problem P without the integrality constraints.

Solution 3: The optimal value of the objective function in $R(P)$ is always less than or equal to the optimal value of the objective function of P .

Solution 4: If the optimal solution of $R(P)$ is integer, it is also an optimal solution of P .

5 Exact methods for discrete optimization

Question 17

An integer minimization problem (IP) is solved with branch and bound. The linear relaxation (LP) is used for bounding. Which one of the following statements is **wrong** ?

Solution 1: By branching, the optimal value of the created subproblems increases, if it exists.

Solution 2: The objective function of each feasible solution of (LP) is an upper bound of (IP).

Solution 3: After branching, the relaxation problem might have an infeasible solution.

Solution 4: If the upper bound in one iteration is less than the lower bound, the branch (node) is discarded.

Question 18

Suppose that we are using the Branch & Bound algorithm to solve the minimization problem P , with the objective function $f(x_1, x_2)$. In one of the iterations, we have calculated the global minimum of the relaxation of the subproblem P_k as $\ell(P_k) = 50.3$, in point $(x_1, x_2) = (13, 7.8)$. The current upper bound is $f^* = 50$. Based on this information, how will the current iteration end?

Solution 1: The global minimum of P is found and the algorithm is stopped.

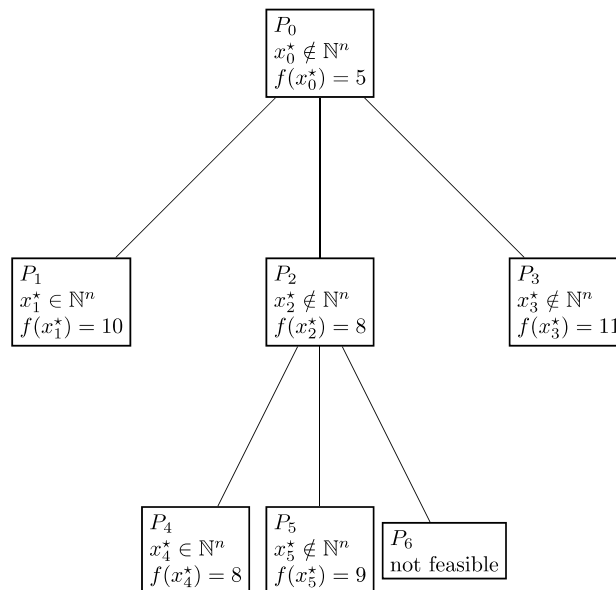
Solution 2: Subproblem P_k is further branched on variable x_1 .

Solution 3: Subproblem P_k is further branched on variable x_2 .

Solution 4: Subproblem P_k is discarded from the Branch & Bound tree.

Question 19

We obtain the following tree for which x_i^* is the optimal solution for the relaxation problem P_i , when solving an integer linear minimization problem by the Branch & Bound algorithm.



Which statement is **correct**?

Solution 1: The value of the optimal solution is 10.

Solution 2: A lower bound for the optimal solution value is 10.

Solution 3: The value of the optimal solution is 8.

Solution 4: There are still nodes in the tree that need to be explored.

Question 20

Consider the following integer optimization problem:

$$\begin{aligned} & \min -x_1 - x_2 \\ \text{s.t:} \\ & 2x_1 + 5x_2 \leq 20, \\ & 4x_1 + 3x_2 \leq 17, \\ & x_1, x_2 \in \mathbb{N}. \end{aligned}$$

We solve the problem using the Gomory cuts algorithm. In the first iteration, we obtain the following optimal tableau of the relaxation.

x_1	x_2	x_3	x_4	
0	1	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{23}{7}$
1	0	$-\frac{3}{14}$	$\frac{5}{14}$	$\frac{25}{14}$
0	0	$\frac{1}{14}$	$\frac{3}{14}$	$\frac{71}{14}$

Which one of the following inequalities is a valid inequality that can be generated at this step?

Solution 1: $x_2 \leq 3$.

Solution 2: $x_1 \leq 1$.

Solution 3: $x_2 - x_4 \leq 3$.

Solution 4: $x_1 + x_3 \leq 1$.